DATA ARCHITECTURE DEEP-DIVE • Distributed SQL Engine

Apache Trino Essentials & PySpark Integration

Unlocking interactive sub-second analytics, data virtualization, and Lakehouse synergy across distributed storage.

PRIMARY ENGINE
Trino (PrestoSQL)

ETL & ML PARTNER
PySpark Engine

STORAGE LAYER
Iceberg / Delta Lake

EXECUTIVE OVERVIEW

What is Apache Trino?

Distributed SQL Engine

An open-source, MPP SQL query engine designed to query petabytes of data directly in place without data movement.

- ANSI SQL Compliant Standard
- Zero Storage Ownership Model
- Pipelined In-Memory Streams

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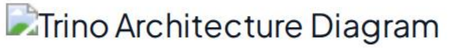
Interactive query response target

Originally PrestoSQL

Created at Facebook to conquer slow Hive MapReduce queries on massive datasets.

SYSTEM INTERNALS

Master-Worker Cluster Architecture



1. Coordinator Node

Parses SQL, creates query plans, and manages active worker scheduling.

2. Worker Nodes

Execute query stages in memory and stream intermediate results via REST APIs.

3. Pluggable Connectors

Abstract physical storage into standard tables, push down filters, and prune splits.

MODULAR DESIGN

Four Key Structural Components

Coordinator

Brain of the cluster.

Workers

Parallel execution muscle.

Connectors

Storage abstraction layer.

Catalogs

Schema & table mounts.

EXECUTION FLOW

Life of a Trino SQL Query

STEP 01
Client Submit
ANSI SQL query received by Coordinator.

STEP 02
Logical Plan
Parser validates schema and generates cost plan.

STEP 03
Stage Split
Work divided into parallel execution splits.

STEP 04
In-Memory Stream
Workers stream data without writing to disk.

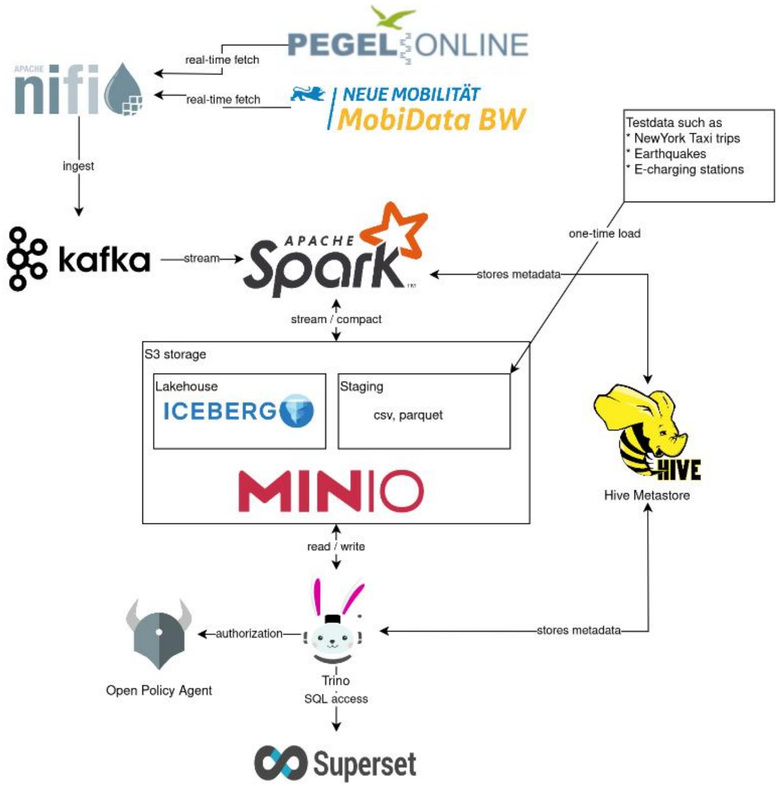
STRATEGIC POSITIONING

Why Adopt Trino alongside PySpark?

Dimension	Apache Trino	PySpark Engine
Primary Use Case	Interactive SQL & BI Dashboards	Heavy Batch ETL & Machine Learning
Query Latency	**Sub-second to Seconds**	Minutes to Hours
Cluster Overhead	Always-on, zero JVM spin-up delay	Dynamic driver & executor provisioning
Data Federation	Cross-source joins in single SQL statement	Manual multi-dataframe connection logic

INTEGRATION PATTERN A

Shared Storage: The Apache Iceberg Synergy



1. PySpark Writes

Performs heavy ETL, cleans raw telemetry, and commits Parquet files into Iceberg tables.

2. Shared Catalog

AWS Glue or REST catalog manages atomic snapshot metadata for both engines.

3. Trino Serves

Instantly serves live BI tools and ad-hoc analysts with sub-second SQL responses.

INTEGRATION PATTERN B

Direct Querying via PySpark JDBC Driver

```
# Reading Trino federated tables directly into PySpark
trino_df = (
    spark.read.format("jdbc")
    .option("url", "jdbc:trino://trino-coord:8080/catalog/schema")
    .option("dbtable", "aggregated_user_metrics")
    .option("user", "pyspark_job")
    .option("driver", "io.trino.jdbc.TrinoDriver")
    .load()
)

# Convert directly to PySpark ML Pipeline
trino_df.createOrReplaceTempView("feature_store")
```

Data Pushdown

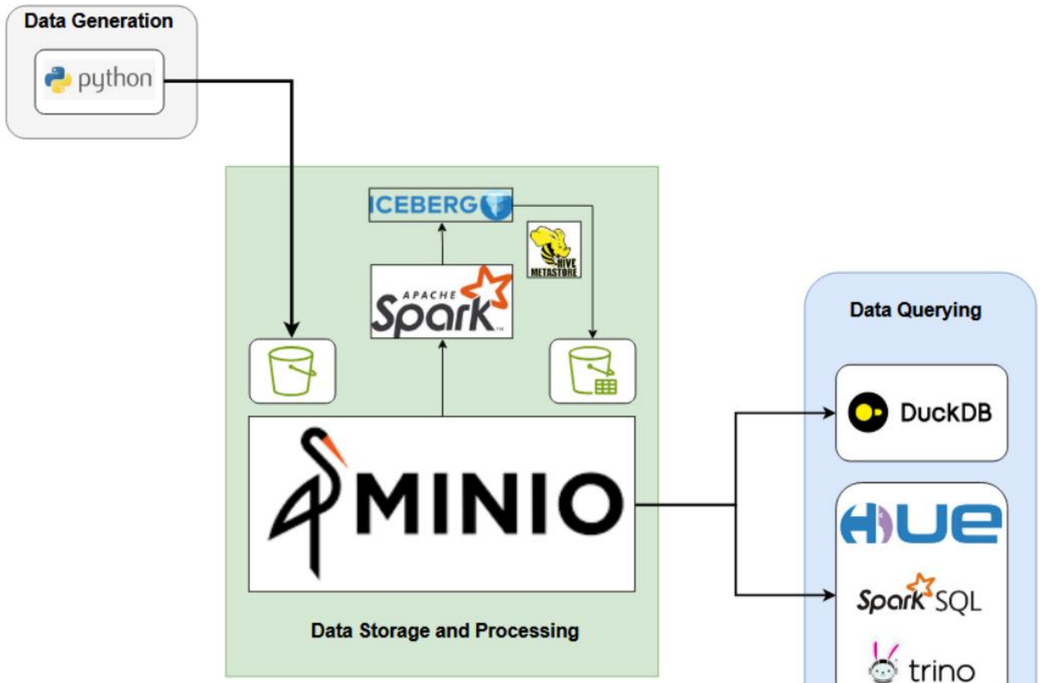
Filter predicates executed inside Trino workers before JDBC transfer.

Federated Fetch

PySpark reads joined multi-database datasets in a single step.

MODERN DATA STACK

End-to-End Lakehouse Architecture



Storage Layer

S3, GCS, or MinIO storing Parquet & ORC formats.

Compute Layer Split

PySpark for batch pipelines; Trino for ad-hoc SQL.

Consumption Layer

Tableau, Power BI, and Jupyter Notebooks.

PERFORMANCE TUNING

Trino Optimization Techniques

Predicate Pushdown

Pushes SQL WHERE clauses down into connector storage engines.

Cost-Based Optimizer

Uses table statistics to reorder joins and optimize memory allocation.

Partition Pruning

Skips unnecessary directories and files during scan phase.

DEPLOYMENT GUIDELINES

Production Architecture Recommendations

COMPUTE SEPARATION

Dedicated Clusters

Isolate BI user clusters from heavy scheduled ETL worker nodes.

GOVERNANCE

Apache Ranger Plugin

Enforce row and column level security policies across all data sources.

MEMORY SPILLING

Disk Spilling Support

Enable task disk spilling to prevent Out-Of-Memory (OOM) query failures.

SUMMARY & SYNERGY

Trino + PySpark = Modern Lakehouse

Use PySpark to transform heavy batch data. Use Trino to deliver sub-second SQL performance.

Questions & Discussion

Image Sources



https://www.nitorinfotech.com/wp-content/uploads/2025/09/architecture-of-apache-trino-1024×531.webp

Source: www.nitorinfotech.com



https://docs.stackable.tech/home/stable/demos/_images/data-lakehouse-iceberg-trino-spark/overview.png

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